

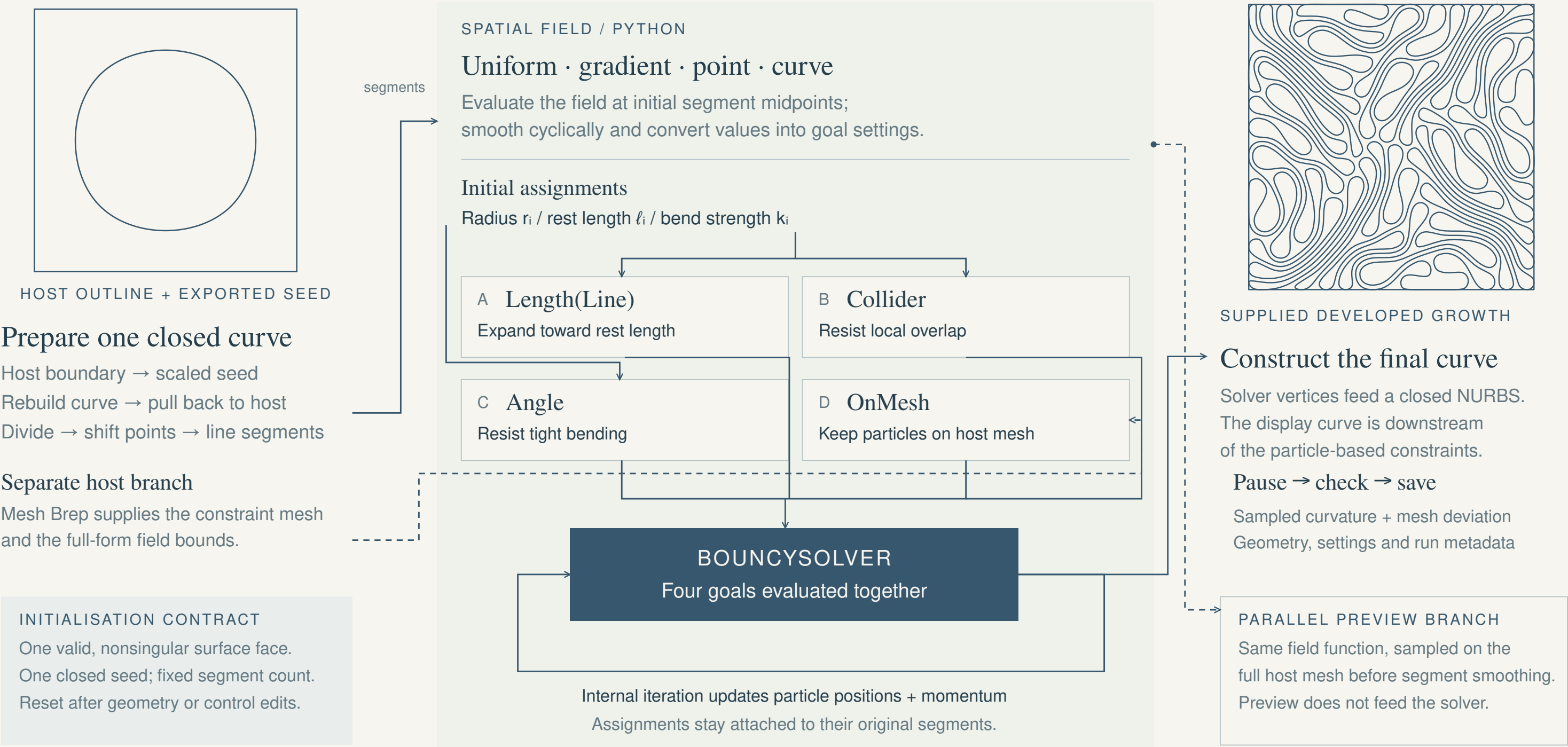
A controllable growth system

Host geometry defines the domain. Local assignments steer the goals. Iterative relaxation develops the curve.

01 / INITIALISE

02 / ASSIGN + SOLVE

03 / READ THE RESULT

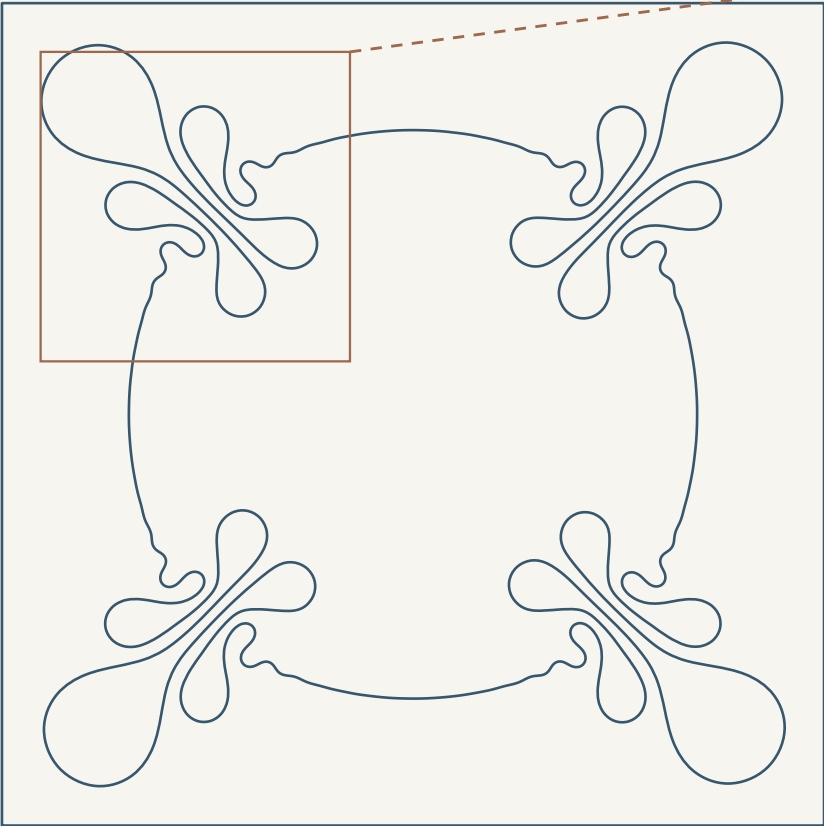


READING THE SYSTEM Solid: geometry / goal data Dashed: host / preview branches Loop: solver iteration

Anatomy of controlled growth

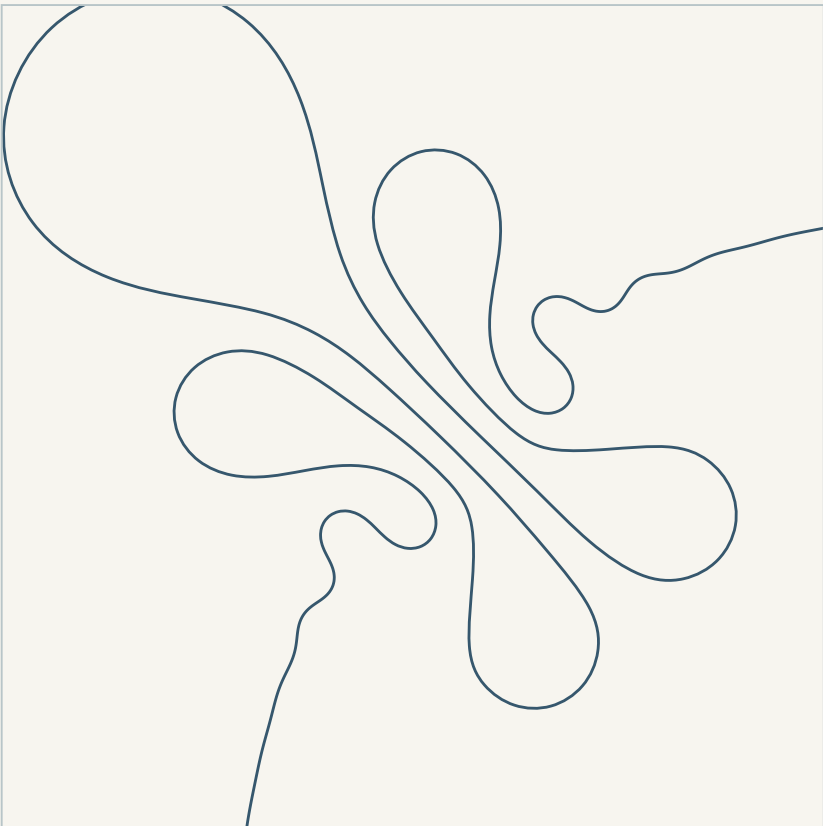
A fixed segment network balances expansion, local separation, bending resistance and a shared host constraint.

01 / GEOMETRY AS EVIDENCE



EXPORTED EMERGENT FOLDING
Actual source curves, retained in full.
This drawing is an output state,
not a plot of solver particles or forces.

02 / LOCAL STRUCTURE



Enlarged window into the same vector paths.
More available length can form folds as goals compete.
Collision and bending goals influence spacing and curvature.

03 / FOUR COUPLED GOALS

- A Expansion**
Length goals seek prescribed rest lengths.
Net expansion requires target total length to exceed the initial total length.
- B Separation**
Collider acts on the segment network.
The assigned radius controls local clearance but does not certify an intersection-free result.
- C Bending**
Angle goals resist changes in direction.
Lower assigned stiffness permits tighter folds; it is not an exact loop-diameter target.
- D Conformity**
OnMesh constrains particles to the host mesh.
The final smooth curve is constructed afterward; its mesh deviation is checked separately.

04 / DENSITY BECOMES THREE LOCAL SETTINGS

$$d_i = C^\beta \bar{f}_i$$

C: contrast · β : blend
 $\bar{f}_i$: smoothed initial field value
 $\beta = 0$ gives the scalar baseline.

Spacing

$$r_i = r_0 \text{ clamp}(d_i^{-s}, 0.3, 3)$$

Dense assignments reduce radius.

s: spacing contrast

Expansion

$$\ell_i = \ell_0 \text{ clamp}(d_i^g, 0.65, 1.5)$$

Packing boost adjusts rest length.

g: local packing / expansion

Bending

$$k_i = k_0 \text{ clamp}(j_i^{-2q}, 0.12, 6)$$

$j_i = \sqrt{(d_i \ d'_i)}$, using paired lines.

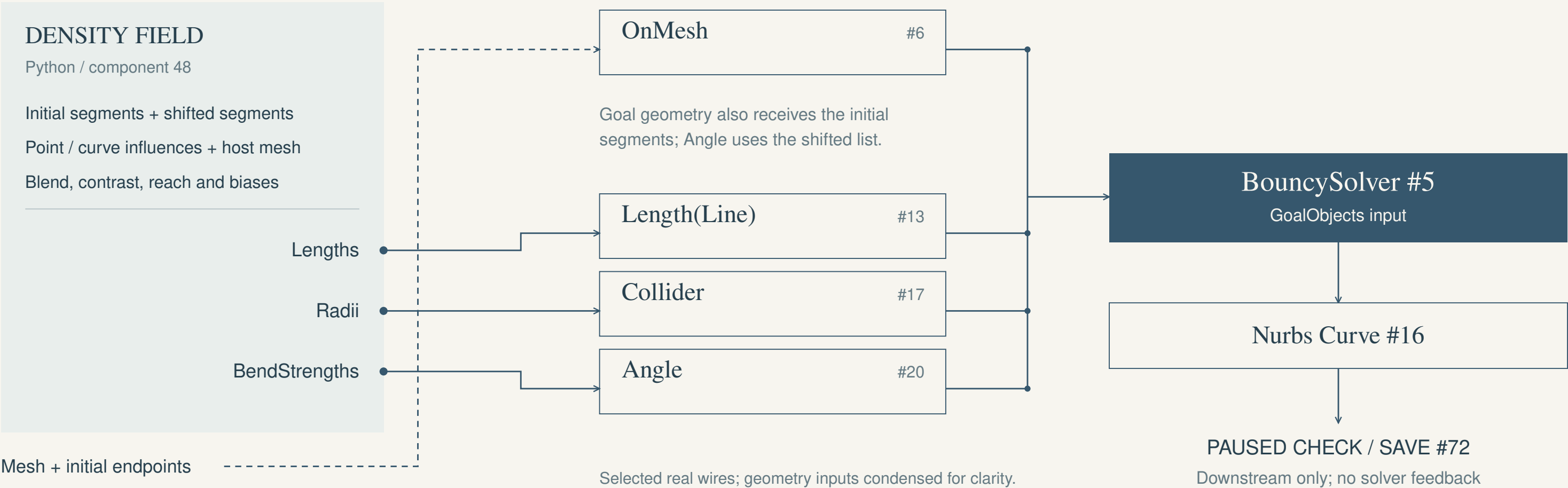
q: loop-size contrast; d'_i : paired assignment

From definition to evidence

Traceable components, explicit controls and a separate verification branch make the system inspectable.

01 / IMPLEMENTED DENSITY-TO-SOLVER CONNECTIONS

PORT NAMES FROM THE SAVED DEFINITION



02 / SAVED CONTROL STATE

Seed segments / scale	1500 / 0.50
Rest length / collision radius	10.000 / 1.127
Density blend / contrast	0.00 / 2.00
Bend / expansion / collision strength	3000 / 10 / 200
Momentum / iterations per update	0.60 / 30

Saved values are not verified settings for every supplied export.

03 / ENGINEERING DECISIONS

- Repair seed placement
Rebuild first, then PullToBrepFace before division.
- Keep the growth network fixed
Density modifies goal settings; no adaptive insertion.
- Separate diagnostics from simulation
Paused sampling + export never alter the solver state.

04 / EVIDENCE BOUNDARY

- Verified from the definition:
component connections, scripts,
saved values and host reference.
- Checks sample 1000 intervals.
Coverage and overlaps are not tested.
- Native Rhino execution was not run
during this documentation audit.